An Automatic Ship Detection Method Adapting to Different Satellites SAR Images With Feature Alignment and Compensation Loss

Siyuan Zhao, Zenghui Zhang, *Senior Member, IEEE*, Weiwei Guo, *Member, IEEE*, and Ying Luo

***Abstract*— Traditional deep learning Synthetic Aperture Radar (SAR) image object detection methods fail to provide effective detection results when faced with SAR image datasets with different joint probability distributions obtained from multiple imaging satellites. In this article, an automatic SAR image object detection method based on domain adaptation is proposed to adapt to unlabeled target domain datasets acquired by different satellites. On the basis of introducing an adversarial domain adaptation learning strategy, we propose Adversarial Learning Attention (ALA) and Compensation Loss Module (CLM) on the baseline network. In ALA, considering the great difference in the scattering intensity of SAR images, the entropy vector can be used to distinguish the high-entropy and low-entropy regions among them and assign the corresponding weights, based on which the adversarial domain adaptation learning attention is proposed to achieve instance-level feature alignment and pixel-level feature alignment in source domain and target domain, respectively. In CLM, the domain alignment of pixel-level feature and instance-level feature of SAR image objects is first implemented, and then to make the feature alignment of both domains more accurate, better aggregation of proposals of different classes of prototype objects in the same domain is required, and further compensation loss is proposed to further restrict the prototype alignment of SAR object features in both domains. We conduct experiments on two SAR datasets obtained from different satellites whose results show the superiority of the proposed method over the state-of-the-art (SOTA) methods and the effectiveness of the proposed module in improving the detection accuracy.**

***Index Terms*— Adversarial learning attention (ALA), compensation loss, deep learning, domain adaptation, synthetic aperture radar (SAR).**

I. Introduction

As an important active remote sensor, synthetic aperture radar (SAR) has become an important observation equipment due to its ability to actively transmit and receive electromagnetic waves and its penetration capability in complex weather condition [1]–[3]. SAR image object detection is a fundamental task of SAR application, which is to extract and analyze effective object information from a large amount of SAR imagery data, so as to achieve the purpose of ground and sea surface information perception [4]–[6].

Constant False Alarm Rate (CFAR) algorithm considers that SAR image object has strong scattering features, and the background clutter conforms to a specific distribution [7], [8]. Although CFAR improves the detection speed compared to traditional manual interpretation, the detection accuracy for complex sea scenes is not high. Since 2006, deep learning has gained worldwide attention for its high prediction accuracy on large and high-dimensional datasets [9], [10]. Detectors based on deep learning are mainly divided into one-stage detection algorithms and two-stage detection algorithms. One-stage detection algorithms have obvious advantages over the two-stage algorithms in terms of training speed, but it is obviously insufficient in accuracy [11]. As a representative of two-stage detection algorithms, Faster Region Convolutional Neural Network (RCNN) uses a trainable regional proposal network to replace the optional search methods of RCNN and Fast RCNN, improving detection accuracy and speed [12].

In recent years, with the rapid development of satellite-borne SAR imaging platforms, an increasing number of high-quality SAR datasets have emerged [13]. Researchers also have begun to put forward a variety of advanced algorithms for SAR image object detection which are gradually moving toward the direction of automatic, fast, and accurate processing of SAR image datasets [14]–[17]. As more and more labeled SAR datasets become available, many classical deep learning methods have achieved good detection results [18], [19].

Both training and test data must be obtained from the same probability distribution in traditional supervised deep learning. The model learns by training with a labeled dataset and then achieving satisfactory test results on the same dataset under this condition [20]. More practically, however, training and test data are collected in different joint probability distributions in many fields [21]–[23]. Current training and test datasets may have significantly different joint probability distributions due to the differences in radar parameters, imaging modes, views, scenes, and other characteristics, especially in the field of SAR image object detection.

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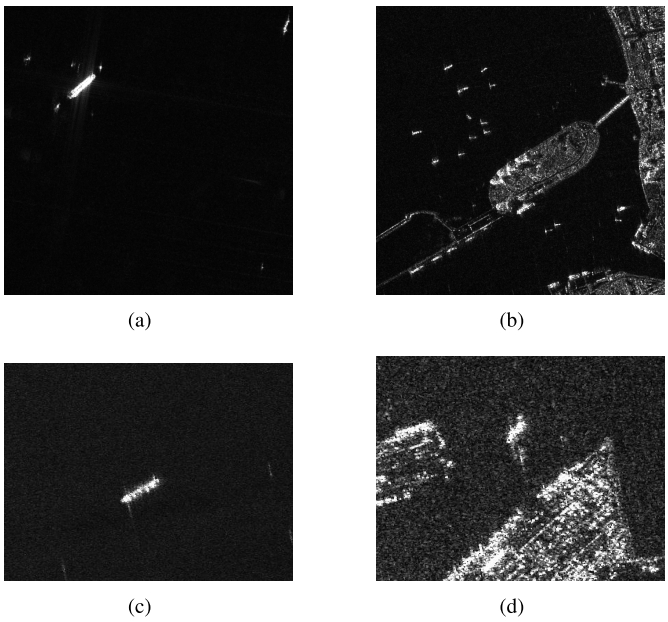


Fig. 1. Gaofen-3 and SSDD SAR images ship detection datasets in similar scenes. (a) Single-ship scene and (b) inshore multi-ship scene in Gaofen-3 dataset. (c) Single-ship scene and (d) inshore multi-ship scene in SSDD dataset.

In our article, we use the following two SAR ship detection benchmark datasets Gaofen-3 and SSDD [18] as experimental datasets, which come from distinct satellites. As shown in Fig. 1, the similar scenes in the two datasets are substantially different in terms of imaging quality, object profile, etc. The image resolution of Gaofen-3 is much better than that of SSDD for single-ship imaging scenes in both datasets, where the ship imaging results in SSDD show more sidelobe interference and the influence of sea clutter is more obvious. The image scene in Gaofen-3 is substantially more intricate for the inshore scenes in the second column, in addition to the impact of resolution and sea clutter. The more specific details of both datasets are presented in Section IV. Therefore, these two datasets imaged from different satellites clearly have different joint probability distributions. When the dataset to be tested is unlabeled and has different joint probability distributions with another labeled dataset, the previous deep learning methods cannot achieve good detection results.

To train better models for adapting to unlabeled dataset with different distributions, it is common practice to collect another batch of data samples for training in the new test dataset to replace or supplement the original training data. However, re-collecting and labeling data will be costly in term of human and time-intensive quantities. Therefore, how to use the existing training data to obtain a better test result on the newfound but unlabeled test data becomes a bottleneck. Recently, machine learning community has proposed domain adaptation to address the above problem, with a focus on finding effective mechanisms to transfer knowledge from one domain to another [24]–[27]. Domain adaptation object detection aims to learn a robust object detector using label-rich data in source domain and label-poor data in target domain. In domain adaptation problem, source domain and target domain come from datasets with different joint probability distributions, which is particularly common in SAR image object detection community [28], [29].

Unlike image classification (image-level class prediction) or semantic segmentation (pixel-level class prediction) [30], object detection task involves bounding box localization and the bounding box class prediction task. Since training and test data are usually assumed to come from the same distribution in previous object detection studies, Chen *et al.* [24] address domain transfer at two levels: 1) image-level transfer, such as image style, lighting, etc., and 2) instance-level transfer, such as object color, size, etc., with the goal of improving the cross-domain robustness of object detection. Image-level represents the encoder feature output, instance-level represents the Region Proposal Network (RPN) [31] feature output, and pixel-level represents pixel points in the image. Zhuang *et al.* [25] achieve image-level and instance-level feature alignment by training adversarial domain classifiers to roughly align in a hierarchy-nested approach and by employing deep semantic information and fine-grained instance representations to establish strong relationships between classes and domains. Instance alignment can have a positive impact only if the detector becomes relatively stable in source domain and target domain. However, due to the large differences in the distribution of source domain and target domain data, there is a large gap between the feature spaces of the jointly trained detector networks, which leads to poor test performance. Compared with the feature alignment method, domain adaptation detection method based on image-to-image conversion can reduce the domain gap of the input level. Chen *et al.* [26] combine CycleGAN [32] and image-to-image conversion to improve the feature discriminability of detector, which is called Harmonizing Transferability Calibration Network (HTCN). HTCN uses image-to-image conversion module to create interpolated images which assist HTCN to fill in the distribution gap between domains, thereby reducing the deviation of the decision boundary. Pseudo-label, as an effective way in semi-supervised learning, is the operation of adding predictive labels to unlabeled data when training a model [33]. Considering that pseudo-label generated by instance-level feature may be incorrect and using pseudo-label as a naive method for the real-case degrades' performance, Kim *et al.* [27] introduce a weak self-training method and adversarial background score regularization for domain adaptation object detection.

Generally, most existing domain adaptation studies have been conducted in the natural image or computer vision communities, but comparable study is rarely found in SAR image object detection community. Shi *et al.* [34] propose a domain adaptation SAR object detection method which achieves good detection results on SAR vehicle datasets from airborne SAR by combining the above-mentioned CycleGAN and pseudo-label methods. Although the experimental results verify the effectiveness of the proposed method, it is important to consider that SAR images are grayscale images so that CycleGAN can only align source domain and target domain at the image-level. Simultaneously, the introduction of CycleGAN to construct a "Diverse Domain" will complicate network parameter adjustment and raise training

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costs. The precision (PR) of the pseudo-label has a significant impact on detection accuracy, and a bad pseudo-label can jeopardize the training operation. As previously stated, while dealing with unlabeled SAR images object detection from different satellites, an appropriate solution at the level of evaluating SAR image object features must be presented.

To detect unlabeled target domain SAR image ship objects from different satellites, an automatic SAR images detection method based on domain adaptation with feature alignment and compensation loss is proposed in this article. We recap the $\mathcal{H}$-distance metric induced by the gap between source domain distributions and target domain distributions [35], which leads to the adversarial domain adaptation learning strategy. On the basis of Faster RCNN as the baseline network, we propose two modules to implement domain adaptation SAR image object detection: 1) adversarial learning attention (ALA) and 2) compensation loss module (CLM).

Since SAR images are grayscale images, unlike natural images where color features can be considered, we need to focus more on achieving domain feature alignment such as edges, contours, textures, and sizes. Therefore, ALA proposes the adversarial domain adaptation attention by means of the entropy vector obtained from the feature vector by GRL and gradient detach strategy [36]. Since the scattering intensity of each region in the SAR image varies greatly, the introduction of ALA based on entropy vectors makes the SAR image features better focused on the detected object. Thus we compute feature vectors in the form of entropy vectors to distinguish high-entropy regions from low-entropy regions in SAR images. Then, we fuse the entropy vector into the obtained feature vector so that the high-entropy regions are given higher weights and the low-entropy regions are given lower weights, thus achieving better adversarial domain adaptation learning.

In CLM, contextual instance-level vectors (feature vectors associated with SAR image ship objects) and the final feature vector are integrated into an overall feature vector, and then domain alignment between pixel-level features and instance-level features is achieved through adversarial domain adaptation learning attention. For SAR ship scene images, the images contain the sea, the ships, and the complex coast scenes. Since different classes of instance features usually contain multi-modal information in SAR image object detection scenes, it is difficult to achieve feature alignment in source domain and target domain only by adversarial domain adaptation learning. To align feature vectors such as edges, contours, textures, and sizes of SAR image objects, we will extract more precise information about each feature vector from the region proposals, thus allowing more embedding clustering of proposals describing the same class of objects. Considering that sample-rich classes should have a high confidence level in the embedding clustering phase of extracting proposals, while two spatially closer proposals are more likely to depict the same object and should be assigned higher connection weights. We then select the value with the highest confidence in the proposed clusters to compute the connection weights for that class and propose a domain adaptation prototype alignment compensation loss to align the different classes of features from both domains.

In a nutshell, the main contributions of this article can be summarized as follows.

1) ALA constructs a novel adversarial domain adaptation attention that assigns corresponding computational weights to high- and low-entropy regions in SAR images by computing entropy vectors and fusing them into feature vectors to achieve better instance-level feature alignment of both domains at the feature level of edges, contours, textures, sizes, etc.
2) To allow for more embedding clustering of proposals describing different classes of objects, enabling the detector to better distinguish different classes of instance-level features, CLM proposes a domain adaptation prototype alignment compensation loss, which in turn enables more accurate feature alignment of source domain and target domain.
3) We conduct extensive experiments on SAR datasets obtained from different satellites. Experimental results clearly show that the proposed method surpasses the state-of-the-art (SOTA) domain adaptation methods for SAR image object detection from different satellites and the proposed modules are both effective for improving the detection accuracy.

The remainder of this article is organized as follows. Section II recaps $\mathcal{H}$-distance of adversarial domain adaptation learning. Section III introduces the proposed novel domain adaptation framework for SAR image object detection. Experimental results and corresponding analysis based on the Gaofen-3 and SSDD measured datasets are given in Section IV. Finally, Section V concludes this article.

II. $\mathcal{H}$-Distance of Adversarial Domain Adaptation Learning

This section recaps a $\mathcal{H}$-distance metric, which is used to introduce the adversarial domain adaptation learning technique. For writing convenience, source domain dataset can be defined as $\mathcal{S} = \{x_i^{\mathcal{S}}, b_i^{\mathcal{S}}, y_i^{\mathcal{S}}\}_{i=1}^{N_{\mathcal{S}}}$, where $x_i^{\mathcal{S}}$ is ith image in $N_{\mathcal{S}}$ fully labeled SAR images and $b_i^{\mathcal{S}}$ denotes the bounding box annotations with corresponding class labels $y_i^{\mathcal{S}}$ of $x_i^{\mathcal{S}}$. In addition, each class label denotes one of the K objects in the dataset and an additional class of the background class, i.e., $y_i^{\mathcal{S}} \in \{1, 2, \ldots, K+1\}$. Furthermore, we also define the target domain dataset as $\mathcal{T} = \{x_i^{\mathcal{T}}\}_{i=1}^{N_{\mathcal{T}}}$, where $x_i^{\mathcal{T}}$ is ith image in $N_{\mathcal{T}}$ unlabeled SAR images with different joint probability distributions from source domain. Inspired by Ben-David *et al.* [35], we need to consider measuring the differences between two sets of samples with different joint probability distributions. The $\mathcal{H}$-distance between $\mathcal{S}$ and $\mathcal{T}$ can be defined as the following form:

$$d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{S},\mathcal{T}) = 2 \sup_{\substack{h\in\mathcal{H} \\ h'\in\Delta\mathcal{H}}} \left|\Pr_{x\sim\mathcal{S}}\left[h(x)\neq h'(x)\right] - \Pr_{x\sim\mathcal{T}}\left[h(x)\neq h'(x)\right]\right| \quad (1)$$

where $h(x): x \to \{0,1\}$ and $h'(x): x \to \{0,1\}$ are classification functions. $\mathcal{H}\Delta\mathcal{H}$ is symmetric difference hypothesis space for

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two different hypothesis spaces $\mathcal{H}$ and $\Delta\mathcal{H}$. Specifically, this $\mathcal{H}$-distance means that the function $h(x)$ makes the probability of $\Pr_{x\sim\mathcal{S}}[h(x) \neq h'(x)]$ as large as possible and the probability of $\Pr_{x\sim\mathcal{T}}[h(x) \neq h'(x)]$ as small as possible in this space. $h(x)$ can also be thought of as a function that is utilized to identify the source and target domains in adversarial domain adaptation learning.

Since the source data are labeled, to estimate the divergence of the distribution of the unlabeled data, the upper bound of the target error is introduced which minimizes the combined error of both domains as [35]

$$\mathrm{Err}_{\mathcal{T}}\left(h, h'\right) \leq \mathrm{Err}_{\mathcal{S}}\left(h, h'\right) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{S}, \mathcal{T}) \tag{2}$$

where $\mathrm{Err}_{\mathcal{S}}$ and $\mathrm{Err}_{\mathcal{T}}$ denote the expected prediction error of samples in source domain and target domain, respectively.

We can note that the upper bound of the target domain error is determined by (2), on which the domain adaptation problems all depend, focusing on minimizing the upper bound by reducing the domain difference between source domain and target domain. Specifically, the adversarial domain adaptation learning strategy involves minimizing the upper bound given in (2) by directly minimizing the $\mathcal{H}$-distance. This distance is inversely proportional to the error of the domain classification. The goal in domain adaptation problem is to reduce this distance, that is, to increase the domain classification error.

Ganin and Lempitsky [37] propose a gradient reversal method to train the adversarial learning neural network model according to (2). The overall goal of the method is to achieve the domain-invariant feature space of CNN with the help of the domain discriminator. The gradient reversal method often appears between the feature extractor F and the domain discriminator D. Generally, the feature extractor F tries to increase the domain classification loss and reduce detection loss in the overall training pipeline. Conversely, domain discriminator D does the opposite. This is embodied as an adversarial loss, which can be written as

$$\max_{F} \min_{D} \{E_{\mathcal{S}}(F, D) + E_{\mathcal{T}}(F, D)\} \tag{3}$$

where E_S and E_T denote the expected domain classification error over source domain and target domain, respectively. With the help of (3), gradient reversal layer (GRL) is proposed to deploy between the feature extractor F and the domain discriminator D to achieve adversarial domain adaptation learning of labeled source domain and unlabeled target domain. The update of GRL is very similar to the stochastic gradient descent (SGD) update of the feed-forward depth model, where GRL has no parameters associated with it (except for the meta-parameter, which are not updated by back propagation). During forward propagation, GRL acts as an identity transform. During back propagation, however, GRL takes the gradient from the subsequent layer, multiplies it, and passes it to the previous layer.

From an empirical point of view, it may be difficult to directly calculate the $\mathcal{H}$-distance of both domains. Therefore, we propose domain discriminators to approximate it in the low, middle, and high layers of the backbone network, respectively. Specifically, domain discriminator attempts to distinguish which domain the feature gradient comes from.

III. Proposed Domain Adaptation Method

In this section, the details of proposed method are presented. The source domain and target domain are represented in the same way as in Section II. The goal of the proposed method is to train an automatic detector which can perform well on unlabeled SAR image in target domain. Specifically, Fig. 2 presents the overall domain adaptation detection framework.

We select Faster RCNN as the baseline network for the proposed domain adaptation detector. In Faster RCNN, the whole image is input to CNN for multilayer feature extraction. RPN is used to produce a series of anchor boxes which need to be cropped and filtered, and then determine whether the anchor boxes belong to the object by softmax. Meanwhile, another branch of bounding box regression corrects the anchor boxes to form more accurate proposals.

Although scene layouts, object co-presence, and backgrounds may vary from source domain to target domain, the description of the same object across domains should be consistent and matchable at the feature level, such as edge, contour, texture, and size. We can assume that these features of each region in SAR image are also the determinants of the domain-invariant features in the domain adaptation task, and these local regions are more descriptive and dominant than other regions [38]. Multiple pixel-level features alignment and instance-level features alignment of SAR objects in source domain and target domain are achieved in ALA by introducing the proposed adversarial domain adaptation learning attention to the low, middle, and high layers of the backbone network, respectively. Because the scattering intensity of regions in SAR ship images varies greatly, the entropy of homogeneous regions is lower than that of heterogeneous regions; for example, the entropy of inshore scenes is higher than that of other offshore scenes, so introducing an entropy vector-based ALA improves the focus of SAR image features on detected objects. To accomplish feature alignment of SAR images at multiple pixel-level, we use GRL to implement adversarial domain adaptation learning on the low, middle, and high layers of the backbone network, respectively. Furthermore, the relationship between entropy and receptive field is addressed in the convolution calculation to better align both domains at the feature level, and the generated feature vectors are fed to domain discriminator to further compute the appropriate entropy vectors. According to Shannon's entropy characteristics, less uncertainty (low-entropy) means higher information purity and vice versa. The acquired entropy vectors are then fused back into feature vectors to achieve the differentiation between high- and low-entropy regions in SAR images and assign associated weights, allowing for better feature alignment between both domains. Over-alignment of source and target features across the dataset can lead to negative transfer due to differences in cross-domain context, object co-occurrence, and scene layout. However, most leading domain adaptation detection methods treat the target domain as a whole, without considering the

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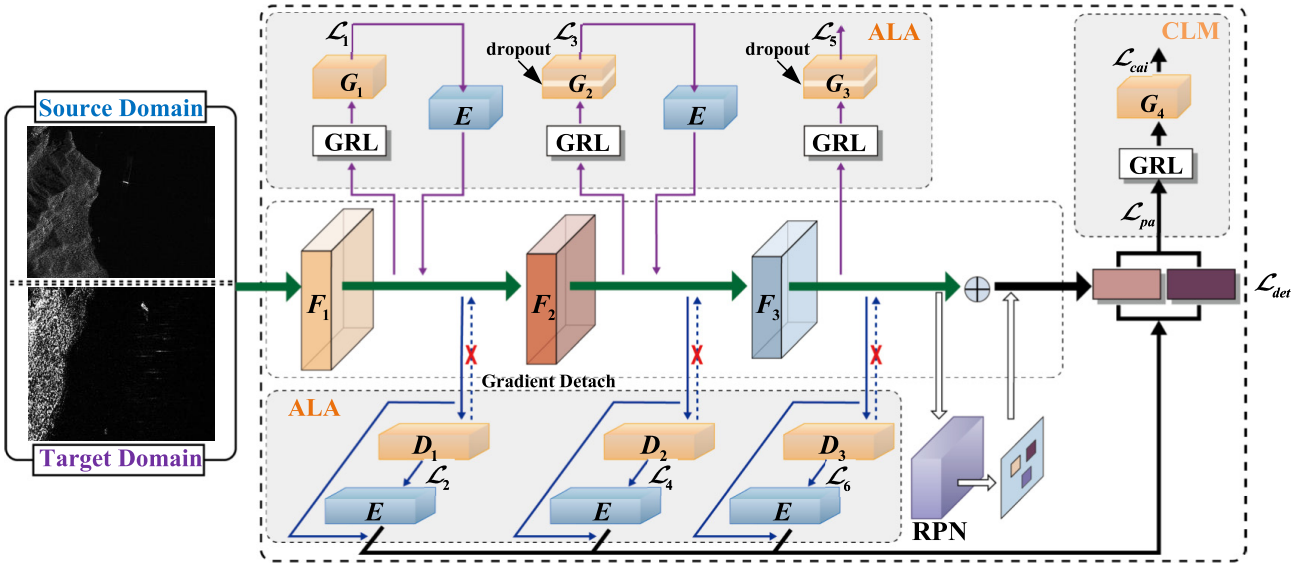


Fig. 2. Automatic ship detection framework adapting to different satellites. We use two SAR datasets from different satellites as source domain and target domain. GRL is gradient reversal layer, G_1–G_4 and D_1–D_3 are domain discriminators, F_1–F_3 are feature extractors in baseline network, and E is calculating process of entropy vector. ALA and CLM are two modules we proposed for domain adaptation detection method.

structure of the data distribution within the class. For SAR image objects, different samples in the same class may have different adaptation difficulties, so we consider reassigning the weights. In addition, because the parameters of SAR object are highly sensitive, to learn them adaptively, we add "dropout" into the domain discriminator introduced in the middle layer and high layer of the backbone network [39]. The gradient detach technique is employed to better extract the contextual information of the feature vectors to obtain better alignment of SAR image ship objects at instance-level. Contextual information is any and all information that may affect the scene and the way objects within it are perceived [40], which is applied to the contextual instance-level vector, but gradient back propagation affects the feature extraction in the backbone network. To achieve the differentiation between high-entropy and low-entropy regions, we fuse the entropy vector acquired by domain discriminator into the extracted feature vector as the contextual instance-level vector for relevant layers in the same way as discussed previously. The instance-level vectors for the low, middle, and high layers are combined into a new instance-level vector that will be used to align with pixel-level features.

In CLM, our goal is to align pixel-level features and instance-level features from source domain and target domain, and better aggregate features of prototype objects in different classes in the same domain. Previous studies have confirmed that instance-level alignment is making use of the region of interest (ROI)-level representations [41]–[43]. Moreover, Yosinski *et al.* [44] have studied the deep features that must eventually transfer from domain-agnostic to domain-specific. Besides, the context instance-level vectors we have obtained are aggregated from the low, middle, and high layers of the network have the features of relative invariance across domains. As a result, the conflict between the transferability and discernability of domain-invariant features can be well resolved if we reasonably integrate pixel-level and instance-level features to accomplish feature alignment between both of them. RPN generates instance proposals in Faster RCNN to discriminate between foreground and background. Due to the deviation of the bounding box, the instance proposals are often distributed around the real object, which leads to the incomplete information of representing the object with a single proposal. Two proposals that are spatially close to each other are more likely to depict the same class, hence they should be given a greater connection weight. In the meantime, because the instance features of distinct classes in SAR image object detection often contain multi-modal information, feature alignment across source domain and target domain is challenging to achieve just by adversarial domain adaptation learning. Thus, while aligning feature vectors such edges, contours, textures, and sizes of SAR image objects, we want to extract more precise information about each feature vector from region proposals, so that proposals portraying distinct classes of objects have more embedding clusters. The connection weight for this class is then calculated using the value with the highest confidence in the proposed clustering, and a domain adaptation prototype compensation loss is proposed to align the identical class features from both domains. The following three parts will go over the detailed technical specifics.

A. Adversarial Learning Attention

Technically, an input SAR image $x_i \in X$ is first fed into the proposed network. The feature map $\{f_1^{H\times W}\}_i = F_1(x_i)$ can be thereby obtained via feature extractor F_1, where H and W denote height and width of the feature map, respectively, while we omit the number of channels for simplicity. Then, the above feature map is fed into the SAR object domain discriminator

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G_1 after GRL which we have mentioned above to achieve adversarial domain adaptation learning. Then we can obtain feature-wise probability output $g_1^{H\times W} = G_1(f_1)$. Thus, the objective function of low-layer adversarial domain adaptation training can be written as follows:

$$\mathcal{L}_1 = \frac{1}{N_{\mathcal{S}} HW}\sum_{i=1}^{N_{\mathcal{S}}}\sum_{j}\log\Big(G_1\Big(\big(\{f_1\}_i^{\mathcal{S}}\big)_j\Big)\Big)^2 + \frac{1}{N_{\mathcal{T}} HW}\sum_{i=1}^{N_{\mathcal{T}}}\sum_{j}\log\Big(1-G_1\Big(\big(\{f_1\}_i^{\mathcal{T}}\big)_j\Big)\Big)^2 \quad (4)$$

where $(\{f_1\}_i^{\mathcal{S}})_j$ and $(\{f_1\}_i^{\mathcal{T}})_j$ denote the jth element of the feature maps from the ith source domain and target domain inputs, respectively. $\mathcal{L}_1$ stabilizes the training of domain classifiers to align low-layer features, which aims to align each perceptual domain of a feature with another domain. The corresponding entropy vector is then calculated as follows:

$$e_1 = E\big(g_1^{H\times W}\big) = -g_1^{H\times W}\odot\log\big(g_1^{H\times W}\big) \quad (5)$$

where $\odot$ stands for Hadamard product. Due to the principle of convolution, we extract each element on the entropy feature map to a local region corresponding to a receptive field on the SAR image. Therefore, we can distinguish between low-entropy region and high-entropy region in SAR image. We fuse the entropy vector into the feature vector as the input to the next feature extractor

$$w_1 = f_1\odot(1+e_1). \quad (6)$$

This will assign lower weight to the low-entropy region of SAR image, which means they will have less uncertainty. Otherwise, the high-entropy region of an image with higher uncertainty is assigned higher weight.

On multi-level feature representations such as local areas, images, and instances, contextual instance-level feature alignment can enrich domain invariant information about the object of interest. Specifically, during training, weights at the lower layers are updated according to the gradient of adversarial and detection losses, leading the network to learn invariant features at both higher and lower layers. Therefore, we set feature map vector into another domain discriminator which is consist of three convolution layers, a pooling layer, and softmax. Gradient detach technique can prevent the gradient flow from the adversarial domain adaptation learning attention from passing through the detection baseline network. Such an operation can help to obtain a more discriminative context feature vector. Fig. 2 shows that the path carries diverse information, so for this task, it is superior that the gradient from the path is suppressed, which is achieved by domain discriminator D_1 as follows:

$$p_1 = D_1(\mathrm{GD}(w_1)) \quad (7)$$

where GD denotes gradient detach technique. Therefore, entropy is calculated by the probability output of the (7) as follows:

$$e_2 = -p_1\odot\log(p_1). \quad (8)$$

To assign corresponding weight to specific entropy regions of SAR image, the entropy vector is fused into the feature vector to obtain the contextual instance-level feature vector as follows:

$$w_2 = w_1\odot(1+e_2). \quad (9)$$

These converged feature vectors are calculated separately and semantically guide instance-level aligned domain-invariant feature learning. The loss function of alignment domain-invariant feature training is

$$\mathcal{L}_2 = \frac{1}{N_{\mathcal{S}}}\sum_{I=1}^{N_{\mathcal{S}}}\sum_{j}\log\Big(\big(\{p_1\}_i^{\mathcal{S}}\big)_j\Big)^2 + \frac{1}{N_{\mathcal{T}}}\sum_{i=1}^{N_{\mathcal{T}}}\sum_{j}\log\Big(1--\big(\{p_1\}_i^{\mathcal{T}}\big)_j\Big)^2. \quad (10)$$

If all the weight parameters in the network are learned to update together, then the more parameters are adjusted, the faster the corresponding features will learn and the influence of other features will be diminished. The above phenomenon is called: co-adaptation [39]. Since the proposed domain adaptation detection network is expected to learn the weight parameters adaptively on SAR dataset with small training samples, we add "dropout" [39] to the middle-layer domain discriminator G_2 and the high-layer domain discriminator G_3. Dropout sets a dropout-ratio parameter as the probability that each hidden unit is set to 0. In each training process, different hidden units are randomly set to 0, and only the relevant parameters of the reserved hidden units are updated. Moreover, dropout can be used for both the convolutional layer and the fully connected layer. The random combination process of this network strengthens the generalization ability of the model and better solves the sensitivity of SAR image object parameters. Then we can obtain next feature-wise probability output as follows:

$$g_2 = G_2(f_2) = G_2(F_2(w_1)). \quad (11)$$

The second loss function can thereby be written as follows:

$$\mathcal{L}_3 = \mathcal{C}\big[\log\big(\{g_2\}^{\mathcal{S}}\big)\big] + \mathcal{C}\big[\log\big(1-\{g_2\}^{\mathcal{T}}\big)\big] \quad (12)$$

where $\mathcal{C}$ denotes cross entropy loss function. $\mathcal{L}_3$ measures the performance of a classification model whose output is a probability value, which increases as the predicted probability deviates from the actual label. The entropy vector and the fused feature map vector can be obtained in the same way as (5) and (6)

$$e_3 = -g_2\odot\log(g_2) \quad (13)$$

$$w_3 = f_2\odot(1+e_3). \quad (14)$$

In addition, following the above steps, we deployed another adversarial domain adaptation learning attention in the middle layer with the same computational steps as (7)–(10)

$$p_2 = D_2(\mathrm{GD}(w_3)) \quad (15)$$

$$e_4 = -p_2\odot\log(p_2) \quad (16)$$

$$w_4 = w_3\odot(1+e_4) \quad (17)$$

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$$\mathcal{L}_4 = \frac{1}{N_{\mathcal{S}}}\sum_{i=1}^{N_{\mathcal{S}}}\sum_j \log\Big(\big(\{p_2\}_i^{\mathcal{S}}\big)_j\Big)^2 + \frac{1}{N_{\mathcal{T}}}\sum_{i=1}^{N_{\mathcal{T}}}\sum_j \log\Big(\big(\{p_1\}_i^{\mathcal{T}}\big)_j\Big)^2. \tag{18}$$

Because we have extracted the relevant features of the SAR image in the low layer and adaptively adjusted the parameters in the middle layer, we use the full connection layer in the high-layer domain discriminator G_3 of the network, so that the previously extracted features and parameters can be fully mapped to the final detection process. Therefore, the third loss function of adversarial domain adaptation training is written as follows:

$$\mathcal{L}_5 = \mathcal{F}\big[\log\big(G_3\big(\{f_3\}^{\mathcal{S}}\big)\big)\big] + \mathcal{F}\big[\log\big(1 - G_3\big(\{f_3\}^{\mathcal{T}}\big)\big)\big] \tag{19}$$

where $\mathcal{F}$ denotes focal loss function [45] which puts more weight on hard-to-classify examples, thus enabling the assignment of higher weights to classes with sparse samples during domain adaptation training. $f_3 = F_3(w_3)$ is the output of third feature extractor in the backbone network.

Simultaneously, we also deploy the same ALA in the high-layer network as follows:

$$p_3 = D_3(\mathrm{GD}(f_3)) \tag{20}$$

$$e_5 = -p_3 \odot \log(p_3) \tag{21}$$

$$w_5 = f_3 \odot (1 + e_5) \tag{22}$$

$$\mathcal{L}_6 = \frac{1}{N_{\mathcal{S}}}\sum_{i=1}^{N_{\mathcal{S}}}\sum_j \log\Big(\big(\{p_3\}_i^{\mathcal{S}}\big)_j\Big)^2 + \frac{1}{N_{\mathcal{T}}}\sum_{i=1}^{N_{\mathcal{T}}}\sum_j \log\Big(1 - \big(\{p_3\}_i^{\mathcal{T}}\big)_j\Big)^2. \tag{23}$$

Summarily, we propose adversarial domain adaptation learning attention sub-network in the low, middle, and high layers of the network, which can align the object features of SAR images in source domain and target domain at multiple pixel-level and instance-level, respectively.

B. Compensation Loss Module

Technically, those contextual instance-level vectors w_2, w_4 and w_5 are then connected to the final feature vector f_n in the backbone network, resulting in a new instance-level feature vector, represented as

$$\boldsymbol{f}_{\mathrm{cai}} = [w_2, w_4, w_5, f_n] \tag{24}$$

where $[w_2, w_4, w_5, f_n]$ represents the integration of the four vectors in series. This integration strategy is adopted for regularizing the domain discriminator to achieve further feature alignment. The above resulting feature vector is also fed into domain discriminator G_4 after GRL. Especially, in G_4, instead of employing any convolutional layers, we use fully connected layers with the aim of better aligning the feature vector representations, thus helping to reduce the differences in features such as edges, contours, textures, and sizes of SAR objects. Then we can obtain feature-wise probability output $g_{\mathrm{cai}} = G_4(f_{\mathrm{cai}})$. Finally, we can get the loss function of adversarial domain adaptation learning as follows:

$$\mathcal{L}_{\mathrm{cai}} = \frac{1}{N_{\mathcal{S}}}\sum_{i}^{N_{\mathcal{S}}}\sum_j \log\Big(\big(\{g_{\mathrm{cai}}\}_i^{S}\big)_j\Big)^2 + \frac{1}{N_{\mathcal{T}}}\sum_{i}^{N_{\mathcal{T}}}\sum_j \log\Big(1 - \big(\{g_{\mathrm{cai}}\}_i^{T}\big)_j\Big)^2. \tag{25}$$

In addition, we introduce the concept of a connection matrix $\boldsymbol{\Theta} \in \boldsymbol{R}^{N_p \times N_p}$ to describe the relationship between the proposals generated by RPN where N_p is proposals number. Since the feature map of SAR object is aggregated in the corresponding SAR image, we here hope to integrate the multi-modal information contained in different SAR images into the prototype representation. We use the confidence value of the proposal for each class as the connection weight and derive the prototype δ_k as a weighted average embedding of the region proposals as follows:

$$\delta_k = \frac{\sum_{i=1}^{N_p}(\boldsymbol{\Lambda}_{ik}\boldsymbol{\Psi}_{ik})(\boldsymbol{\Lambda}_i\boldsymbol{\Omega}_i)^T}{\sum_{i=1}^{N_p}(\boldsymbol{\Lambda}_{ik}\boldsymbol{\Psi}_{ik})} \tag{26}$$

where $\boldsymbol{\Lambda} = \boldsymbol{A}^{-1/2}\boldsymbol{\Theta}\boldsymbol{A}^{-1/2}$, $\boldsymbol{A} \in \boldsymbol{R}^{N_p \times N_p}$ is a diagonal matrix with elements $A_{ii} = \sum_j \Theta_{ij}$, k denotes class number, $\boldsymbol{\Psi} \in \boldsymbol{R}^{N_p \times a}$ and $\boldsymbol{\Omega} \in \boldsymbol{R}^{N_p \times b}$ (a and b are the number of prototype classes and the dimension of embedding) denote classification confidence matrix and feature embeddings of proposals, respectively. This prototype serves as a substitute for each class in the subsequent domain alignment process.

We also introduce a simple spatial distance to calculate the prototypes between source domain and target domain

$$d\big(\delta_i, \delta'_j\big) = \big\|\delta_i - \delta'_j\big\| \tag{27}$$

where δ_i and δ'_j are in different domains.

Here we define the previously mentioned connection matrix $\boldsymbol{\Theta}$ as the form of Intersection over Union (IoU)

$$\boldsymbol{\Theta}_{i,j} = \frac{r_i \cap r_j}{r_i \cup r_j} \tag{28}$$

where r_i and r_j denote ith and jth region proposals, respectively. Considering that the classes with rich samples are more fully trained and better aligned, especially in the stage of extracting proposals, they should have a high confidence. Based on this fact, we choose the highest confidence of the particular class in a set of proposals, and use this confidence value to calculate the weight of this class

$$c_i = \max_{1 \le k \le N_p}[\boldsymbol{\Lambda}_{ki}\boldsymbol{\Psi}_{ki}]. \tag{29}$$

Furthermore, we propose a prototype alignment loss to limit the distance between prototypes of different classes

$$\mathcal{L}_{\mathrm{pa}} = \frac{\sum_{0 \le i,j}(1 - c_i)\big(1 - c'_j\big)\max\big(0, 1 - d\big(\delta_i, \delta'_j\big)\big)}{\sum_{0 \le i,j}(1 - c_i)\big(1 - c'_j\big)} \tag{30}$$

where c_i and c'_j denote the prototypes of two domains from which pairs of prototypes belonging to different classes are taken. To sum up, it is the compensation loss introduced for prototype alignment.

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C. Overall Training Objective

The overall training objective function which is composed of detection loss and proposed modules loss functions are shown in this section. We employ Faster RCNN and the proposed modules as our overall framework. The detection loss function can be written as follows:

$$\mathcal{L}_{\text{det}} = \begin{cases} \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{rpn}}, & \text{Source Domain} \\ \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}}, & \text{Target Domain} \end{cases} \tag{31}$$

where $\mathcal{L}_{\text{det}}$, $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{reg}}$ and $\mathcal{L}_{\text{rpn}}$ denote object detection task loss, object classification loss, bounding-box regression loss, and RPN loss, respectively. The other modules loss function can be written as follows:

$$\mathcal{L}_{\text{other}} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4 + \mathcal{L}_5 + \mathcal{L}_6 + \mathcal{L}_{\text{cai}} + \mathcal{L}_{\text{pa}}. \tag{32}$$

In domain adaptation problem, the feature extractor and the domain discriminator are more likely playing a rivalry game. The feature extractor would like to extract clearer features to represent the object characteristics, while the domain discriminator needs to "confuse" the feature extractor so that feature classification and domain alignment can be achieved. Therefore, to train the entire model, we summarize the loss function in the above model as the following overall objective function:

$$\max_{G_i, D_i} \min_{F_i} L_{\text{det}} + \lambda L_{\text{other}} \tag{33}$$

where G_i and D_i are ith domain discriminators, F_i denotes ith feature extractors, and λ is a trade-off hyperparameter between detection loss and our proposed modules loss.

IV. Experimental Results and Analysis

In this section, we employ two datasets obtained by diverse satellites to verify the proposed automatic ship detection method based on domain adaptations. Gaofen-3 and SSDD datasets are implemented as source domain and target domain, respectively, to evaluate method which are challenging domain-shift datasets obtained from multiple satellites. Following that, we go over the experimental setting and results in detail.

A. Datasets Description

Gaofen-3 dataset in our experiment contains 1000 images with resolutions from 1 to 3 m. The scene types include harbors, islands, and sea surfaces with different levels of sea state. The objects cover thousands of ships of more than ten classes, such as transport ships, oil tankers, and fishing vessels. Gaofen-3 satellite is a civil microwave remote-sensing imaging satellite in the National High Resolution Earth Observation System major project, and the first C-band multi-polarization high-resolution SAR satellite in China. Gaofen-3 satellite adopts Strip-Map, Fine Strip-Map 1, Full Polarization 1 (QPSI), Full Polarization 2 (QPSII) imaging modes, etc. SSDD dataset proposed by Li *et al.* [18] is imaged by TerraSAR-X, RadarSat-2 and Sentinel-1 satellites, which contains 1160 images with 2456 multi-scale ships. The resolution of these images ranges from 1 to 5 m.

TABLE I
INFORMATION OF GAOFEN-3 AND SSDD DATASETS

Dataset Index	Gaofen-3	SSDD
Imaging satellite	Gaofen-3	TerraSAR-X, Radarsat-2, Sentinel-1
Number of images	1000	1160
Ship classes number	11	8
Average number of ships	3.54 per image	2.12 per image
Resolution	1-3 m	1-5 m

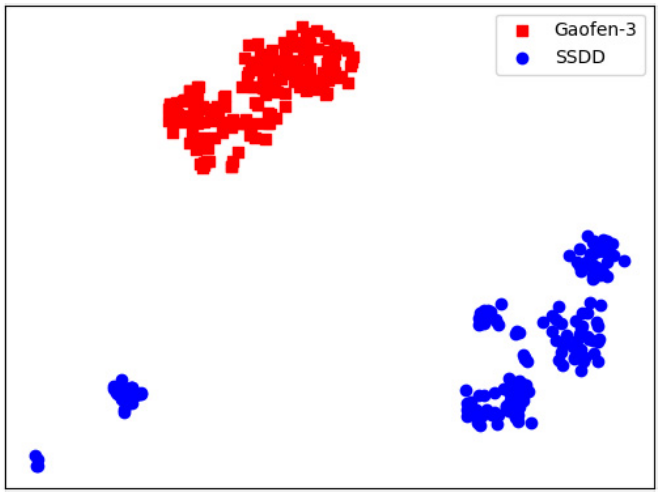


Fig. 3. Two-dimensional t-SNE visualizations from the features of Gaofen-3 and SSDD data samples where red square denotes Gaofen-3 data samples and blue circle denotes SSDD data samples.

Tables I and II present dataset index about the two datasets and the technical parameters of their corresponding imaging satellites. It can be seen that the resolution, incident angle, polarization mode, and imaging mode of the SAR images in both datasets are variant, and the antenna model, attitude control technology, and thermal control technology of the observation satellites are all different. Furthermore, the sea surface clutter probability distributions of the four satellites mentioned above have been shown to obey different statistical distributions through [46]–[49]. However, in the SAR image detection domain adaptation community, the different distributions of source domain and target domain are not simply reflected in the different probability statistical distributions of sea surface clutter, but more in the different distributions of the detected objects in the feature space. We utilize t-distribution stochastic neighbor embedding (t-SNE) [50] to reduce the dimensionality of the object features in both datasets to two dimensions, whose projection result is shown in Fig. 3. As can be seen in Fig. 3, the feature space between Gaofen-3 and SSDD data samples is clearly separable before domain adaptation, which indicates a large domain gap between both datasets. Combining the above presentations of the relevant index and imaging satellites for two datasets, it can be seen that Gaofen-3 dataset and SSDD dataset have different joint probability distributions that satisfy the condition to be source domain and target domain in the domain adaptation problem.

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TABLE II
COMPARISON OF TECHNICAL PARAMETERS BETWEEN GAOFEN-3, TERRASAR-X, RADARSAT-2, AND SENTINEL-1

Technical Parameters	Gaofen-3	TerraSAR-X	Radarsat-2	Sentinel-1
Orbit Altitude	755 km	514 km	798 km	693 km
Peak power	1.5 kW	1.8 kW	1.27 kW	4.7 kW
Antenna size	15 m × 1.5 m	4.8 m × 0.7 m	15 m × 1.4 m	12.3 m × 0.84 m
Incident angle	$10^\circ - 60^\circ$	$20^\circ - 55^\circ$	$20^\circ - 45^\circ$	$10^\circ - 60^\circ$
Elevation sweep angle	$\pm 20^\circ$	$\pm 25^\circ$	$\pm 11^\circ$	$\pm 20^\circ$
Bandwidth	240 MHz	150 MHz	100 MHz	100 MHz
Resolution	0.5-500 m	1-16 m	1-100 m	5-20 m
Imaging swath	10-650 km	5-100 km	20-50 km	20-400 km
Polarization mode	single/double/quad	single/double/quad	single/double/quad	single/double
Lifespan	8 year	5 year	7.5 year	7.5 year
Imaging mode	Spotlight Mode, Strip Mode, Sweep Mode, Wave Mode, QPSI, QPSII	Strip map, Scan, Interference wide range	Strip Mode, TOPS Mode, Wave Mode, Ultra-width Mode	Strip Mode, Sweep Mode, Ultra-fine Mode

B. Experimental Initialization and Parameter Setting

In our experiment, we randomly divide the datasets into a training set and a test set with the proportion of 7:3. Especially, the datasets are not expanded by any expansion method, and no form of label information is provided in the target domain training. In addition, all images are normalized by subtracting the average value from the uniform channel dimension to balance the range of gray values for stable training. Finally, we use a non-maximum suppression (NMS) algorithm [51] to obtain the final detection results. The threshold of NMS and the weight decay rate are set to 0.3 and 0.0001, respectively.

In all experiments, we adopt Faster RCNN as the baseline network and Resnet-101 as backbone network whose weight parameters are pre-trained on ImageNet. We choose SGD with a momentum of 0.9 as the optimization method for the training process. The initial learning rate and λ are set to 1×10^4 and 1, respectively. To evaluate the adaptation performance, IoU threshold is set to 0.5 for all experiments. 0.5 is the empirical IoU threshold common to most existing SAR object detection methods, and in the subsequent ablation experiments, we also give the detection results under different IoU thresholds. We use PR, recall (RE), and mean average PR (mAP) to quantitatively evaluate detection methods, which can be expressed as follows:

$$\mathrm{PR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{34}$$

$$\mathrm{RE} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{35}$$

$$\mathrm{mAP} = \frac{1}{m} \sum_{m} \left[\int \mathrm{PR}(\mathrm{RE}) d(\mathrm{RE}) \right] \tag{36}$$

where TP, FP, and FN denote the number of correctly detected objects, false positives, and false negatives, respectively. PR and RE, respectively, measure the proportion of true positives and the true proportion of true positives in the overall test results. mAP describes the region under the PR–RE curve, which also illustrates the overall performance of the detection method.

All experiments are implemented on a server using Pytorch framework on Centos7 and supported by a Nvidia Quadro RTX 6000 graphics card.

TABLE III
SAR OBJECT DETECTION RESULTS (%) OF VARIOUS METHODS

Method	G → S			S → G		
	PR	RE	mAP	PR	RE	mAP
Faster RCNN [12]	62.3	79.8	66.4	57.2	70.6	57.7
DAF [24]	66.9	80.4	67.8	59.4	72.4	59.8
SAUDA [27]	68.6	81.2	70.4	60.2	76.9	62.8
HTCN [26]	69.4	81.9	72.5	63.6	79.2	64.6
Ours	**74.6**	**82.9**	**78.1**	**69.8**	**80.9**	**68.4**

C. Comparison Experiment With SOTA Methods

To verify the excellent performance of the proposed method, we compare it with four detection methods.

Faster RCNN [12]: This is the baseline of our proposed method and also the basic object detection method in many papers. We chose it mainly to verify the domain adaptation capabilities of the baseline method, and to further promote our research on domain adaptation problems.

Domain Adaptive Faster RCNN for Object Detection in the Wild (DAF) [24]: This is recognized as the first article to start domain adaptation object detection, and it provides ideas for subsequent research.

Self-Training and Adversarial Background Regularization for Unsupervised Domain Adaptive One-Stage Object Detection (SAUDA) [27]: This is the first one-stage domain adaptation object detection method with pseudo labels. We chose it mainly to compare domain adaptation methods with pseudo-labels, and also to compare two-stage and one-stage methods.

HTCN [26]: This is a domain adaptation method using CycleGAN to generate the Interpolation Domain. We chose it mainly to compare with the domain adaptation method that uses CycleGAN to achieve image-to-image conversion.

In Faster RCNN baseline, we only input source domain SAR images. The other three SOTA methods are all unsupervised domain adaptation methods. We define two detection tasks from the source domain to the target domain: Gaofen-3 to SSDD ($G \rightarrow S$) and SSDD to Gaofen-3 ($S \rightarrow G$). Table III, Figs. 4, and 5 are the object detection results of the proposed method and other detection methods on Gaofen-3 dataset and SSDD dataset, respectively.

First of all, we evaluate the object detection results of the test set SAR images in Gaofen-3 dataset and the test set SAR

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images in SSDD dataset, such as the PR, RE, and mAP in Table III. Compared with the baseline network, our proposed method improves mAP by 11.7%. Similarly, compared with SOTA domain adaptation methods, our method surpasses DAF, SAUDA and HTCN by 10.3%, 7.7%, and 5.6% in this task, respectively. For $G \rightarrow S$ detection task, because the scenes in Gaofen-3 dataset are more complex and the ship classes are more diverse, and Gaofen-3 satellite has more imaging and polarizing methods, all metrics in $G \rightarrow S$ are higher than $S \rightarrow G$. However, the proposed method is still 10.7% higher than the baseline network. Compared with other methods, the proposed method still surpasses 8.6%, 5.6%, and 3.8% in this task, respectively. From these indicators and the above analysis, it is clear that our method is superior compared to the SOTA methods.

In $G \rightarrow S$ task, we show experimental results corresponding to four typical ship detection scenes in Fig. 4. For the small and densely distributed ship scene, all the detection methods can accurately detect a large number of ship objects, but there are varying degrees of false alarm or missed detection. In contrast, the proposed method has fewer false alarm and missed detection.

Although all methods accurately detect the docking ship, it can be seen that Faster RCNN detects a ship into three objects. In addition to the methods we proposed, the remaining three methods more or less locate the wrong size of the object box.

Inshore ships detection has always been a difficult point. In the inshore scene, there are a large number of land objects besides ships. These objects have a high scattering coefficient, which has a great influence on the detection of ships. It can be seen that all methods have not achieved the best results, but the proposed method still has the best detection result. Because the proposed domain discriminator can better distinguish between the homogeneous region and the heterogeneous region of SAR images, our method has less false detection of inshore scenes.

There are many false alarms in island scene detection, but our method also has a lower false alarm rate here. The ground truth shows that there is only one ship object in this SAR image, but several methods have more or less false detection. Because the network can only achieve object detection through some specific features, while the ship object and the extended island are similar in size and have similar contour features.

It can be seen from the above results, compared with the baseline network, the domain adaptation detection method has better detection ability when facing domain-shift of the datasets. Because the baseline network does not have an adversarial learning module, making it impossible to learn feature information from one domain to another. Simultaneously, the method based on domain adaptation can constrain the network to learn the common features of the source domain and the target domain by adding adversarial learning module and feature learning module, it can reduce missed detection to varying degrees. Moreover, the proposed method can learn more discriminative features of the target domain and has better results than other comparison methods.

For the domain adaptation detection results in $G \rightarrow S$ task, because the scenes of Gaofen-3 dataset are more complex and ship classes are more plentiful, the target domain detection effectiveness in this task is generally better than the other one. However, in $S \rightarrow G$ task, the detection of the target domain is generally poor due to the relative large number of satellites acquired in SSDD dataset and the poor image quality. Fig. 5 shows the detection results of the four scenes, which all show that our method has fewer false alarm or missed detection. In general, the proposed method has better detection results in these two tasks, which can prove that both modules we proposed are better in the transferability and discriminablity of domain adaptation.

In terms of computational performance, the detailed computation time for each image is shown in Table IV. The proposed method and other domain adaptation detection methods have faster computation time, which is significantly better than the baseline method, and also reflects the better computational performance of the proposed method for SAR images.

D. Ablation Studies

To analyze the different modules of the proposed method, we implement ablation studies using Gaofen-3 dataset and SSDD dataset. By analyzing the results of ablation studies, we can have a more comprehensive understanding of the impact of different modules on detection performance.

1) Ablation Studies of ALA: In this part, we will perform an ablation analysis of ALA. Both tasks were implemented in an ablation study with the same experimental setup as the comparison experiment described above. The experiment includes ablation of the entire ALA and the corresponding domain discriminators. Tables V and VI show the ablation results where "w" denotes that there is the associated domain discriminators.

From the results in Tables V and VI, we can see that mAP with ALA is higher than without ALA. Simultaneously, we compared the detection results of with and without the proposed domain discriminator network branch. To show the results more clearly and not to have too much repetitive experiment steps, we take the low-layer network branch where G_1 and D_1 are, respectively, as a pair of the experiment object. The same process is done for the middle-layer and high-layer network.

Furthermore, we can see $G_1\&D_1$ improves our method the most in both tasks. Since the feature extractors in Faster RCNN have different output dimensions, the features extracted from the lower layer network are the richest, so the domain feature alignment we do on top of them achieves the best results. As the depth of the network increases, fewer features are extracted by the feature extractor and the feature alignment becomes less effective. Therefore, using separate $G_2\&D_2$ in the network yields detection results that are worse than using $G_1\&D_1$ but better than using $G_3\&D_3$. This is also because, as the layers in the network get higher, we extract fewer features and feature alignment becomes more difficult. In Fig. 6, we visualize the heat map of the middle-layer feature map from the ALA module in $G \rightarrow S$ task. Fig. 6 shows that ALA in middle layer is more concerned with heterogeneous regions

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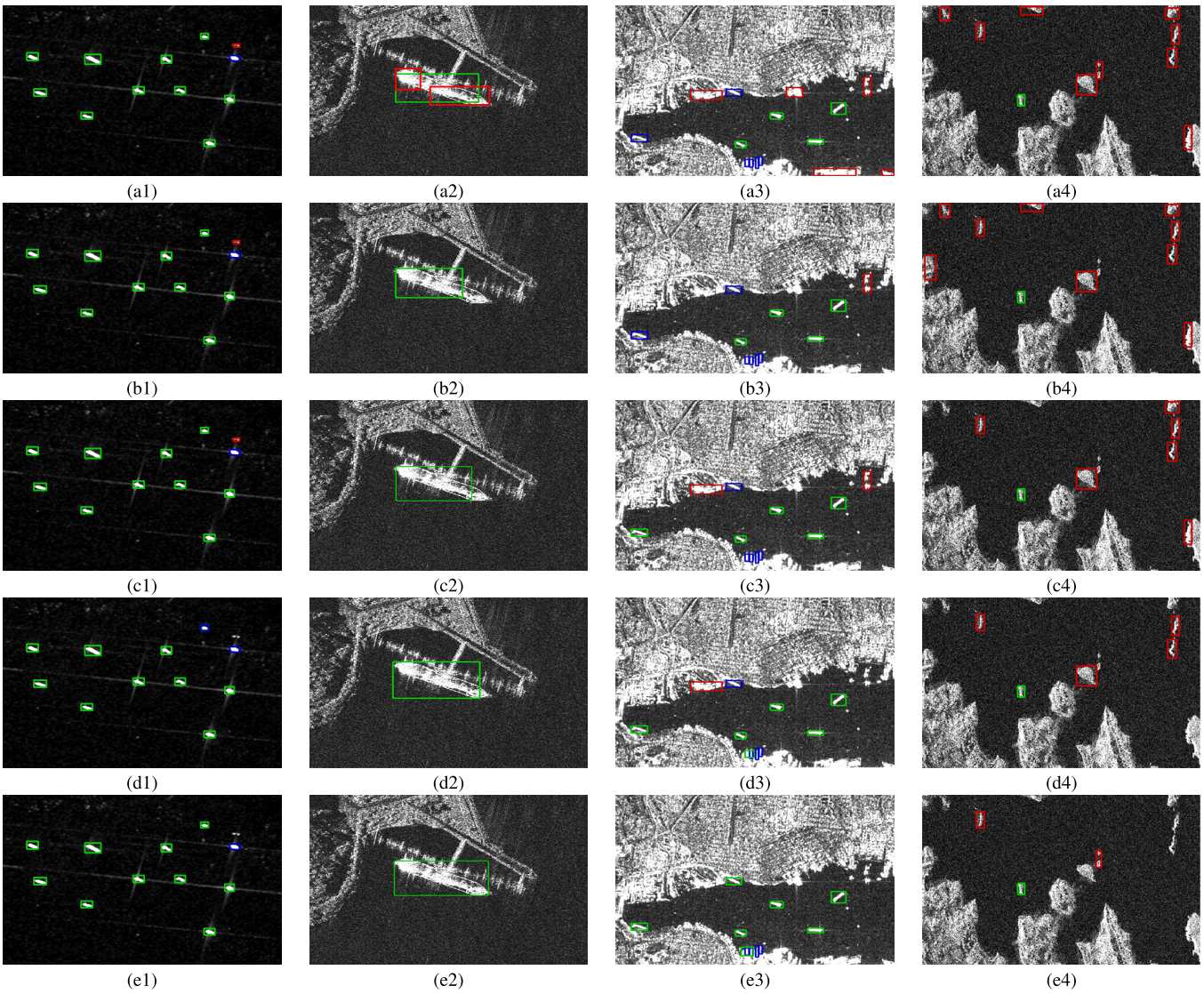


Fig. 4. SAR images tested in SSDD dataset are compared with the intuitive detection results of SOTA methods ($G \rightarrow S$ task). In all the subfigures, the green, red, and blue rectangular box indicate correct detection, false alarm, and missed detection, respectively. The first column is small and densely distributed ships scene, the second column is ship-docking scene, the third column is inshore scene, and the fourth column is island scene. (a1)–(a4) Faster RCNN. (b1)–(b4) DAF. (c1)–(c4) SAUDA. (d1)–(d4) HTCN. (e1)–(e4) Proposed method.

TABLE IV

COMPUTATION TIME OF PER IMAGE FOR VARIOUS METHODS

Method	**Faster RCNN [12]**	**DAF [24]**	**SAUDA [27]**	**HTCN [26]**	**Ours**
G $\rightarrow$ S **time**	209 ms/img	180 ms/img	182 ms/img	**170 ms/img**	171 ms/img
S $\rightarrow$ G **time**	209 ms/img	181 ms/img	184 ms/img	174 ms/img	**172 ms/img**

TABLE V

ABLATION DETECTION RESULTS (%) OF THE PROPOSED ALA AND RELEVANT DOMAIN DISCRIMINATORS IN $G \rightarrow S$

Method	G_1&D_1	G_2&D_2	G_3&D_3	G $\rightarrow$ S
without ALA				69.4
	w			74.0
		w		73.3
with ALA			w	72.4
	w	w	w	**78.1**

TABLE VI

ABLATION DETECTION RESULTS (%) OF THE PROPOSED ALA AND RELEVANT DOMAIN DISCRIMINATORS IN $S \rightarrow G$

Method	G_1&D_1	G_2&D_2	G_3&D_3	S $\rightarrow$ G
without ALA				60.2
	w			59.8
		w		57.6
with ALA			w	56.0
	w	w	w	**68.4**

such as object regions with higher entropy than homogeneous regions such as the background, meaning that SAR objects of interest to ALA are more likely to have high-entropy due to their textures and sizes. The above results prove that our

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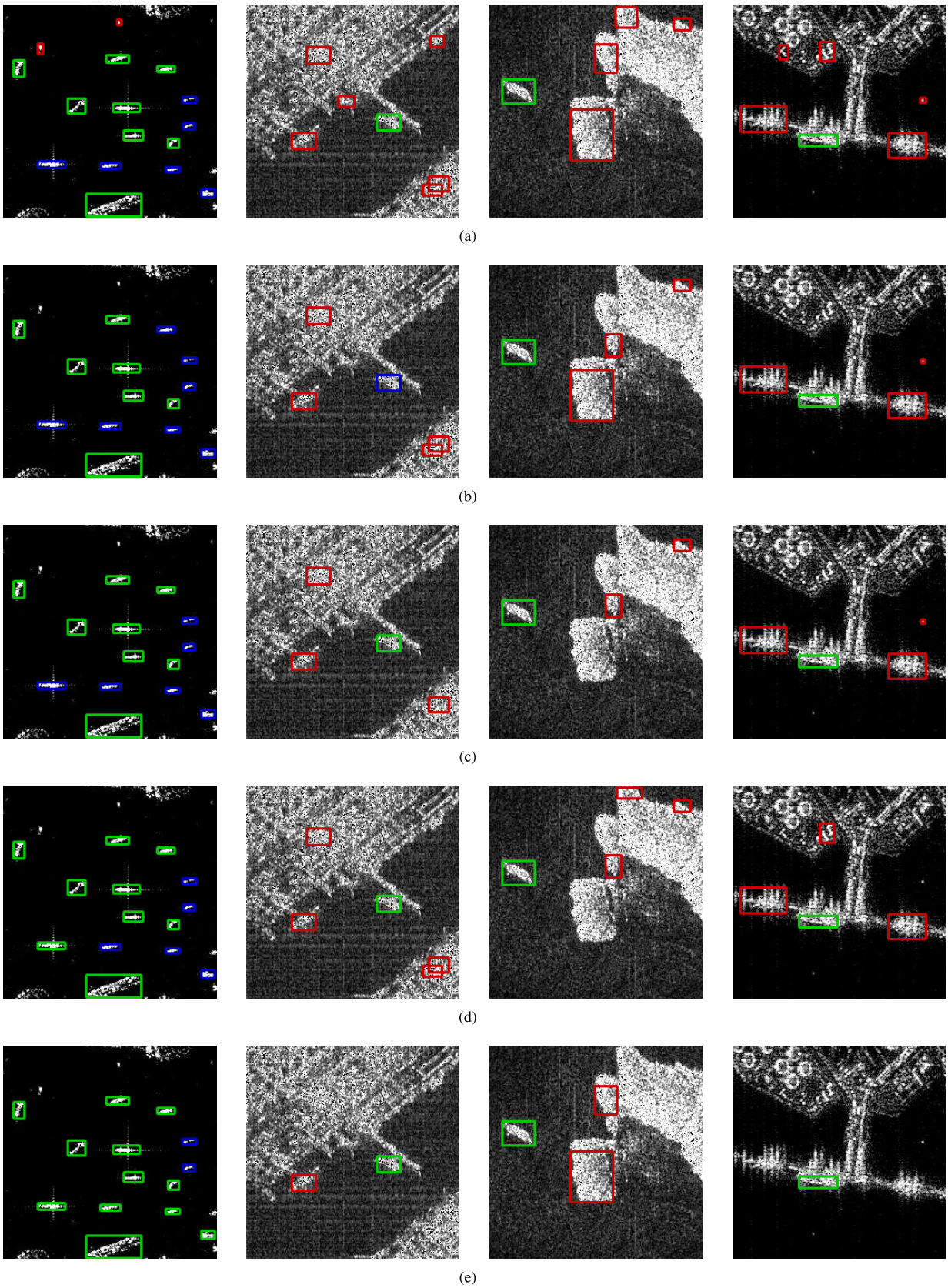


Fig. 5. SAR images tested in Gaofen-3 dataset are compared with the intuitive detection results of SOTA methods ($S \rightarrow G$ task), where the green, red, and blue rectangular box indicate correct detection, false alarm, and missed detection, respectively. (a) Faster RCNN. (b) DAF. (c) SAUDA. (d) HTCN. (e) Proposed method.

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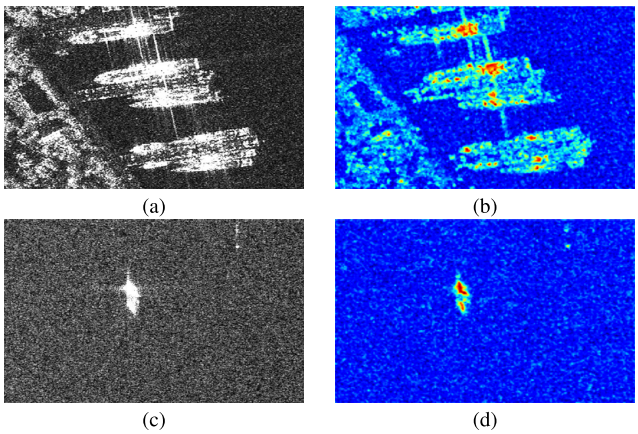


Fig. 6. Heat map of the middle-layer feature map from the ALA module. (b) and (d) are the corresponding heat maps of (a) and (c), respectively.

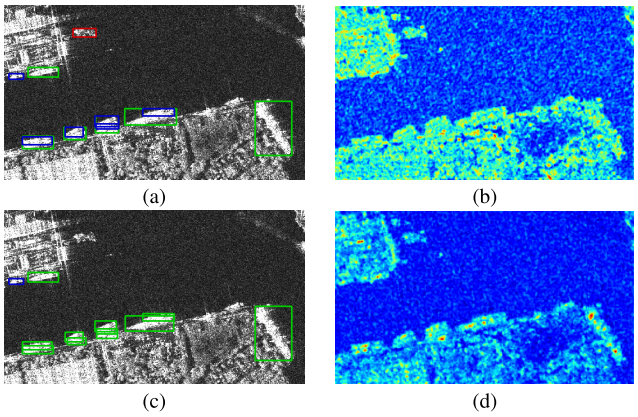


Fig. 7. Detection result of one SAR image in SSDD dataset with and without ALA module, where the green, red, and blue rectangular box indicate correct detection, false alarm, and missed detection, respectively. (a) Detection result and (b) heat map without ALA. (c) Detection result and (d) heat map with ALA.

proposed ALA module can effectively improve the detection results, and the domain discriminator have a better effect when used in the lower network layer than in the higher network layer.

Furthermore, Fig. 7 shows the detection result of one SAR image in SSDD dataset with and without ALA module. It can be seen that the network can detect a certain number of ship objects without ALA module, but misses a certain number of objects compared to the ground truth because of a close distance. Similarly, in the upper portion of this SAR image, one object is mistakenly detected as a ship because it is close to the shore and has a ship-like shape. It can be further seen from its corresponding heat map that the nearby ships do not reflect significant changes in the heat values, and the objects with ship-like shape that appear in the upper portion of the image also have high heat value, which can cause some missed detection and wrong detection by the network. After adding ALA module, it is reflected from the heat map that our method clearly extracts the object region of the ship with high entropy (the region with high heat value). The detection result can also be seen that missed detection and wrong detection significantly reduce after adding ALA.

TABLE VII

ABLATION DETECTION RESULTS (%) OF ENTROPY VECTOR AND GRADIENT DETACH

Method	G → S	S → G
without Entropy Vector	75.6	64.7
replaced by GRL	73.8	63.6
overall proposed network	**78.1**	**68.4**

2) Ablation Studies of Entropy Vector and Gradient Detach: In this part, we first implement the ablation experiment of calculating the entropy vector and fusing into the feature vector to assign the corresponding weight in ALA. Since previous domain adaptation studies do not have such an operation that gradient detach is used instead of GRL in extracting contextual instance-level feature vectors. Therefore, we also conduct the ablation experiment of replacement from gradient separation to GRL. This part of our experiment is divided into two parts in Table VII; the first one shows the results with or without entropy vector and the second one shows the results that gradient detach is replaced by GRL. The experiment results prove that entropy vector can improve the detection results effectively, because it can more clearly distinguish high- and low-entropy regions, and assign them corresponding weights to achieve better feature alignment. Similarly, we found that when we changed gradient detach technique to GRL, the mAP is reduced by 4.3% and 4.8%, which shows that the model based on gradient detach can learn more discriminative features from the context instance-level features.

3) Ablation Studies of CLM: In this section, we conduct ablation experiments on the overall CLM module and then on the prototype alignment compensation loss function in the CLM module. Table VIII shows the detection results of the proposed CLM. We can see that the detection accuracy of the method decreases significantly with the absence of the overall CLM module and $\mathcal{L}_{\text{pa}}$, which indicates that the proposed module and compensation loss are helpful to improve the detection accuracy. Fig. 8 visualizes the clustering results between both domains in $G \rightarrow S$ task by t-SNE. We utilize t-SNE to reduce the dimensionality of the object features in both datasets to two dimensions, which is shown in Fig. 8, where circle and square represent the object in both datasets, respectively. From Fig. 8(a), it can be seen that after feature alignment by the backbone network and ALA module, source domain and target domain have achieved a good clustering alignment result. However, in Fig. 8(b), the objects' features in both domains are more closely distributed and indistinguishable after CLM, which indicates that CLM can result in better alignment of the detected object features in both domains. Finally, the convergence of the entire objective function and the prototype alignment compensation loss function is shown in Fig. 9. In Fig. 9, we can see the proposed $\mathcal{L}_{\text{pa}}$ in both tasks can decrease quickly during the first 2000 iterations and then fluctuate in a small range after 4000 iterations.

4) Ablation Studies of Backbone Network: In all of the above experiments, we use Resnet-101 as backbone network, but there are also studies using Visual Geometry Group (VGG)-16 as the backbone network. Without loss of generality, we conduct the following ablation study to further analyze

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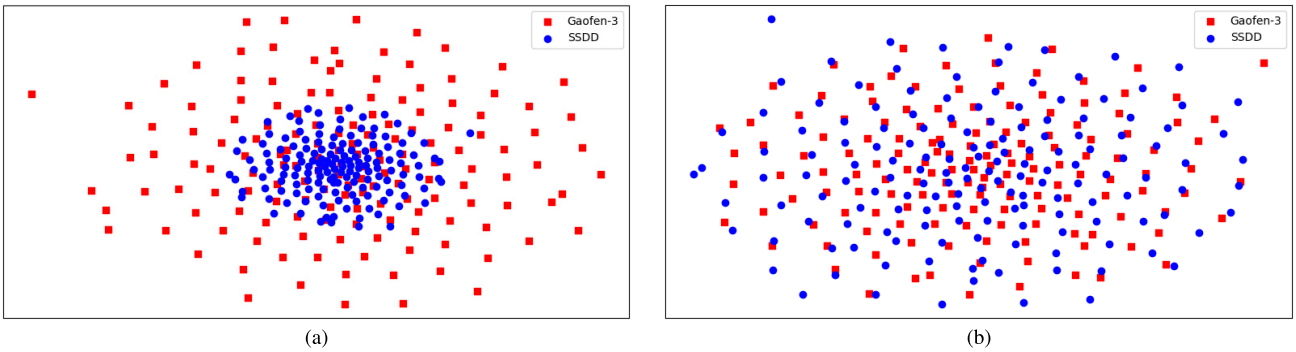


Fig. 8. Visualization of the feature distributions via t-SNE in $G \rightarrow S$ task where red square denotes Gaofen-3 dataset and blue circle denotes SSDD dataset. Feature distributions (a) before and (b) after CLM.

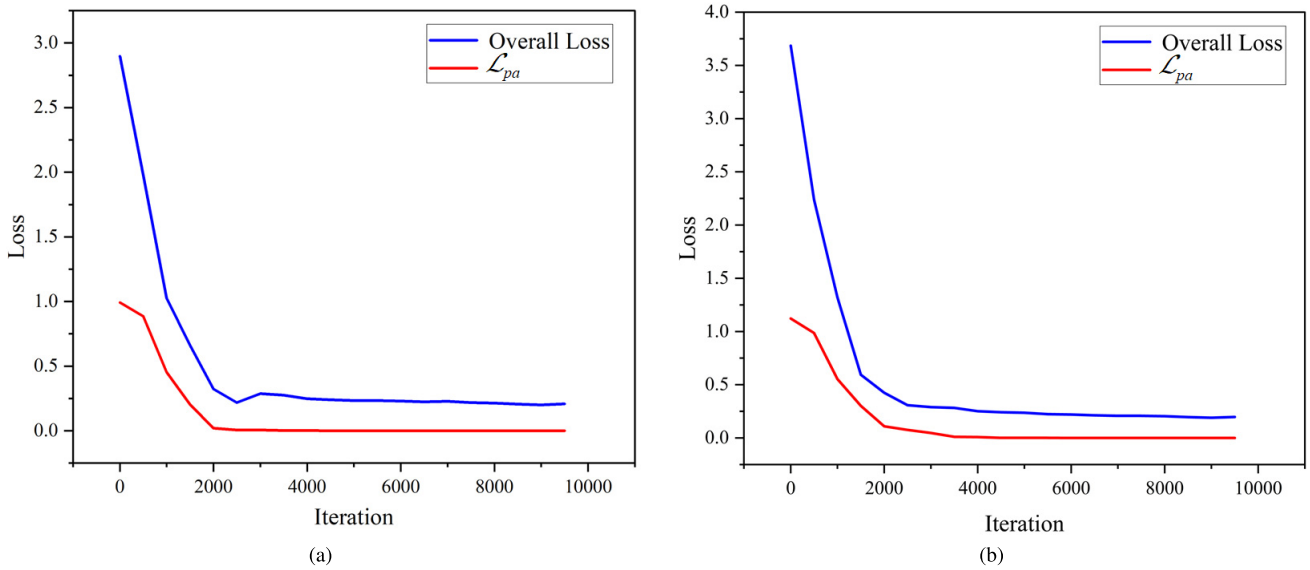


Fig. 9. Convergence curves of overall loss and $\mathcal{L}_{\text{pa}}$ in the proposed CLM module. In both subfigures, blue curve indicates the trend of the overall loss function and the red curve indicates the trend of $\mathcal{L}_{\text{pa}}$. (a) $G \rightarrow S$ task. (b) $S \rightarrow G$ task.

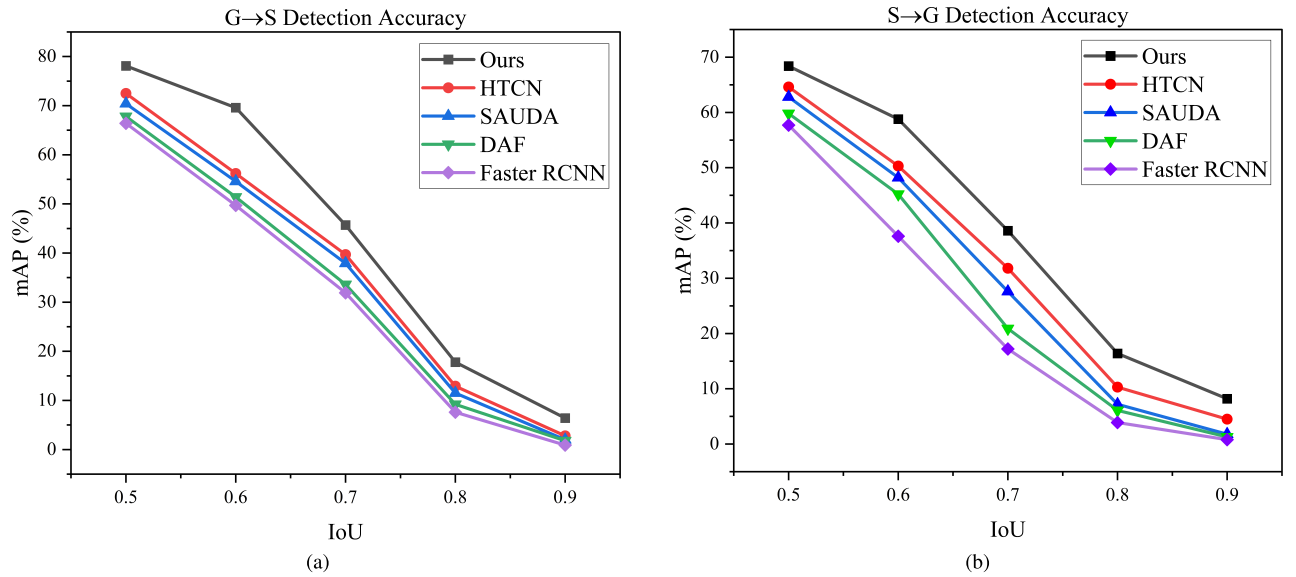


Fig. 10. Detection results with different IoU thresholds. The detection accuracy of two tasks under different IoU thresholds is shown, where purple, green, blue, red, and black denote Faster RCNN, DAF, SAUDA, HTCN, and the proposed method, respectively. (a) $G \rightarrow S$ task. (b) $S \rightarrow G$ task.

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TABLE VIII
ABLATION DETECTION RESULTS (%) OF THE PROPOSED CLM

Method	G → S	S → G
without CLM	71.0	61.6
without $\mathcal{L}_{pa}$	74.6	64.1
overall proposed network	**78.1**	**68.4**

TABLE IX
$G \rightarrow S$ DETECTION ACCURACY (%) OF RESNET-101 AND VGG-16

Method	Resnet-101	VGG-16
Faster RCNN [12]	62.5	54.2
DAF [24]	64.4	58.5
SAUDA [27]	66.8	59.6
HTCN [26]	67.5	62.1
Ours	**78.1**	**69.7**

TABLE X
$S \rightarrow G$ DETECTION ACCURACY (%) OF RESNET-101 AND VGG-16

Method	Resnet-101	VGG-16
Faster RCNN [12]	57.7	52.5.4
DAF [24]	59.8	55.7
SAUDA [27]	62.8	56.6
HTCN [26]	64.6	59.8
Ours	**68.4**	**61.0**

the advantage of the two backbone networks applications in the proposed domain adaptation detection network. Tables IX and X show the detection results of Resnet-101 and VGG-16 based on two tasks, respectively. From two tables, we can clearly see that the detection performance of Resnet-101 is better than that of VGG-16. This is owing to the VGG-16 network's enormous number of parameters that must be learned. However, for domain adaptation object detection, the difficulty lies in how to better extract and align the features in source domain and target domain, and thereby cannot accurately fit a larger number of well-fit parameters during the training process. Resnet-101 network helps to solve the problem of gradient disappearance and gradient explosion, allowing us to train a deeper domain adaptation network while ensuring good performance.

5) Ablation Studies of IoU Threshold: For object detection, IoU threshold is an important factor reflecting the quality of detection. We know from experience that a higher threshold indicates a greater overlap with the ground truth box. In our previous experiments, IoU threshold is set to 0.5. We conduct multiple experiments on different IoU thresholds to better understand the impact of IoU threshold. In Fig. 10, we can see that as the IoU threshold increases, the detection performance of all methods gradually decreases. Under these circumstances, our method still performs better than other methods at different thresholds which further proves the superiority of our proposed method.

6) Ablation Studies of Hyperparameter Sensitivity: The selection of hyperparameter λ in above experiments is based on empirical analysis, and here we have performed a sensitivity analysis on it. λ controls the weight of domain adaptation adversarial feature alignment, whose value is changed in the following experiment. From Table XI, we can see that the mAP of both detection tasks decreases when we increase or decrease the value of this λ. Thus, we select $\lambda = 1$ in our overall experiments.

TABLE XI
HYPERPARAMETER SENSITIVITY ANALYSIS ON BOTH TASKS DETECTION ACCURACY (%)

λ	0.25	0.75	1	1.25	1.75	2
G → S **mAP**	74.3	76.9	**78.1**	77.2	75.4	72.7
S → G **mAP**	66.0	67.3	**68.4**	67.1	65.8	64.5

V. CONCLUSION

In this article, we propose an automatic ship detection method that adapts to different satellites SAR images with feature alignment and compensation loss. We use Faster RCNN as a detection baseline network and develop a domain adaption framework with two modules: ALA and CLM to obtain high-detection accuracy on unlabeled target domain SAR images. $\mathcal{H}$-distance is recapped to describe the difference in the sample space of both domains, which results in the adversarial domain adaptation learning strategy to achieve feature alignment in source domain and target domain. Considering that the scattering intensity of each region in SAR ship images varies greatly, ALA introduces entropy vector-based ALA and assigns corresponding weight to the regions with different entropy, thus achieving multiple pixel-level feature alignment and instance-level feature alignment. While achieving feature alignment between pixel-level and instance-level by adversarial domain adaptation learning attention, CLM describes different classes of objects with more embedded proposals clustering so that the detector can better distinguish between different classes of instance features, and thereby proposes a domain adaptation prototype alignment compensation loss. During the experiments, the labeled source domain and the unlabeled target domain are integrated as the input for training a domain adaptation model which is tested on target domain. Experimental results based on two datasets from different satellites, Gaofen-3 and SSDD, show that the detection accuracy of the proposed method is significantly better than that of the baseline and other SOTA methods, proving the effectiveness and superiority of our method.

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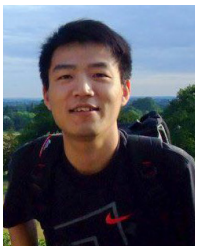
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